COMPARING THE CUTOFF OF A FINITE MODULE WITH THE CUTOFF OF ITS COMPLETION

THE STACKS PROJECT AUTHORS

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Allowed standard prerequisites (editorial).

- The cutoff s_{A,I}(M) of a finite module with respect to an ideal, defined through the depth of localizations at pairs of primes. Defined in the author's section on finiteness of local cohomology, where the counted statement occurs, and recalled at the start of the counted proof. Fixes the two quantities being compared.
- Dimension theory of the completion: comparison of dimensions of localizations of the completion with those of the original ring, and the Ratliff proposition. Both are cross-chapter pointers invoked inside the counted proof. Supplies the inequalities between dimensions of corresponding primes that the comparison of cutoffs is reduced to.

Source reference. Original reference algebra-lemma-apply-grothendieck-module: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2593–2593. Navigation pointer only: the counted text refers to the Stacks item algebra-lemma-apply-grothendieck-module, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Source reference. Original reference more-algebra-lemma-completion-dimension: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 834–834. Navigation pointer only: the counted text refers to the Stacks item more-algebra-lemma-completion-dimension, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

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Source reference. Original reference more-algebra-proposition-ratliff: sources/stacks-local-cohomology/pinned-original-source.tex, source lines 2043–2043. Navigation pointer only: the counted text refers to the Stacks item more-algebra-proposition-ratliff, which belongs to another chapter of the project and is not part of this card. The pointer carries the target's own label as its number because this chapter's staged snapshot does not contain the other chapter's tag table; its content is not reproduced here.

Lemma 0.1. *Let A be a universally catenary Noetherian local ring. Let $I \subset A$ be an ideal. Let M be a finite A-module. Then*

$$s_{A,I}(M) \geq s_{A^\wedge,I^\wedge}(M^\wedge)$$

If the formal fibres of A are (S_n), then $\min(n+1, s_{A,I}(M)) \leq s_{A^\wedge,I^\wedge}(M^\wedge)$.

Proof. Write $X = \mathrm{Spec}(A)$, $X^\wedge = \mathrm{Spec}(A^\wedge)$, $Z = V(I) \subset X$, and $Z^\wedge = V(I^\wedge)$. Let $\mathfrak{p}' \subset \mathfrak{q}' \subset A^\wedge$ be primes with $\mathfrak{p}' \notin Z^\wedge$ and $\mathfrak{q}' \in Z^\wedge$. Let $\mathfrak{p} \subset \mathfrak{q}$ be the corresponding primes of A. Then $\mathfrak{p} \notin Z$ and $\mathfrak{q} \in Z$. Picture

$$\begin{array}{ccccc} \mathfrak{p}' & \longrightarrow & \mathfrak{q}' & \longrightarrow & A^\wedge \\ \downarrow & & \downarrow & & \uparrow \\ \mathfrak{p} & \longrightarrow & \mathfrak{q} & \longrightarrow & A \end{array}$$

Let us write

$$\begin{aligned} a &= \dim(A/\mathfrak{p}) = \dim(A^\wedge/\mathfrak{p}A^\wedge), \\ b &= \dim(A/\mathfrak{q}) = \dim(A^\wedge/\mathfrak{q}A^\wedge), \\ a' &= \dim(A^\wedge/\mathfrak{p}'), \\ b' &= \dim(A^\wedge/\mathfrak{q}') \end{aligned}$$

Equalities by More on Algebra, Lemma more-algebra-lemma-completion-dimension. We also write

$$\begin{aligned} p &= \dim(A^\wedge_{\mathfrak{p}'}/\mathfrak{p}A^\wedge_{\mathfrak{p}'}) = \dim((A^\wedge/\mathfrak{p}A^\wedge)_{\mathfrak{p}'}) \\ q &= \dim(A^\wedge_{\mathfrak{q}'}/\mathfrak{p}A^\wedge_{\mathfrak{q}'}) = \dim((A^\wedge/\mathfrak{q}A^\wedge)_{\mathfrak{q}'}) \end{aligned}$$

Since A is universally catenary we see that $A^\wedge/\mathfrak{p}A^\wedge = (A/\mathfrak{p})^\wedge$ is equidimensional of dimension a (More on Algebra, Proposition more-algebra-proposition-ratliff). Hence $a = a' + p$. Similarly $b = b' + q$. By Algebra, Lemma algebra-lemma-apply-grothendieck-module applied to the flat local ring map $A_{\mathfrak{p}} \to A^\wedge_{\mathfrak{p}'}$ we have

$$\mathrm{depth}(M^\wedge_{\mathfrak{p}'}) = \mathrm{depth}(M_{\mathfrak{p}}) + \mathrm{depth}(A^\wedge_{\mathfrak{p}'}/\mathfrak{p}A^\wedge_{\mathfrak{p}'})$$

The quantity we are minimizing for $s_{A,I}(M)$ is

$$s(\mathfrak{p}, \mathfrak{q}) = \mathrm{depth}(M_{\mathfrak{p}}) + \dim((A/\mathfrak{p})_{\mathfrak{q}}) = \mathrm{depth}(M_{\mathfrak{p}}) + a - b$$

(last equality as A is catenary). The quantity we are minimizing for $s_{A^\wedge,I^\wedge}(M^\wedge)$ is

$$s(\mathfrak{p}', \mathfrak{q}') = \mathrm{depth}(M^\wedge_{\mathfrak{p}'}) + \dim((A^\wedge/\mathfrak{p}')_{\mathfrak{q}'}) = \mathrm{depth}(M^\wedge_{\mathfrak{p}'}) + a' - b'$$

(last equality as $A^\wedge$ is catenary). Now we have enough notation in place to start the proof.

Let $\mathfrak{p} \subset \mathfrak{q} \subset A$ be primes with $\mathfrak{p} \notin Z$ and $\mathfrak{q} \in Z$ such that $s_{A,I}(M) = s(\mathfrak{p}, \mathfrak{q})$. Then we can pick $\mathfrak{q}'$ minimal over $\mathfrak{q}A^\wedge$ and $\mathfrak{p}' \subset \mathfrak{q}'$ minimal over $\mathfrak{p}A^\wedge$ (using going down for $A \to A^\wedge$). Then we have four primes as above with $p = 0$ and $q = 0$. Moreover, we have $\mathrm{depth}(A^\wedge_{\mathfrak{p}'}/\mathfrak{p}A^\wedge_{\mathfrak{p}'}) = 0$ also because $p = 0$. This means that $s(\mathfrak{p}', \mathfrak{q}') = s(\mathfrak{p}, \mathfrak{q})$. Thus we get the first inequality.

Assume that the formal fibres of A are (S_n). Then $\operatorname{depth}(A^\wedge_{\mathfrak{p}'}/\mathfrak{p}A^\wedge_{\mathfrak{p}'}) \geq \min(n,p)$. Hence

$$s(\mathfrak{p}',\mathfrak{q}') \geq s(\mathfrak{p},\mathfrak{q}) + q + \min(n,p) - p \geq s_{A,I}(M) + q + \min(n,p) - p$$

Thus the only way we can get in trouble is if $p > n$. If this happens then

$$\begin{aligned} s(\mathfrak{p}',\mathfrak{q}') &= \operatorname{depth}(M^\wedge_{\mathfrak{p}'}) + \dim((A^\wedge/\mathfrak{p}')_{\mathfrak{q}'}) \\ &= \operatorname{depth}(M_{\mathfrak{p}}) + \operatorname{depth}(A^\wedge_{\mathfrak{p}'}/\mathfrak{p}A^\wedge_{\mathfrak{p}'}) + \dim((A^\wedge/\mathfrak{p}')_{\mathfrak{q}'}) \\ &\geq 0 + n + 1 \end{aligned}$$

because $(A^\wedge/\mathfrak{p}')_{\mathfrak{q}'}$ has at least two primes. This proves the second inequality. $\square$